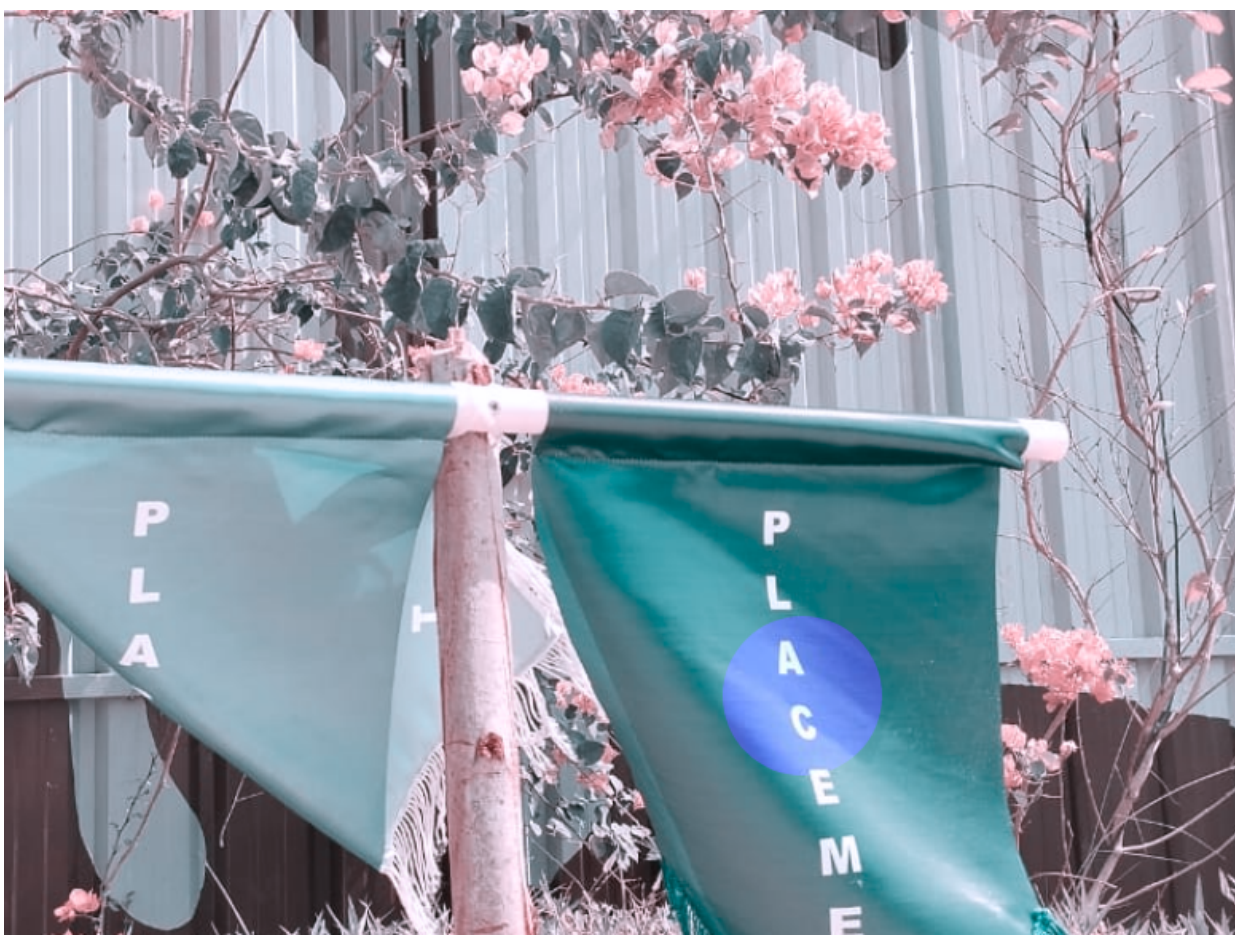


Creating Colourblind Friendly Visualizations for Comprehensible Data Interpretations

COLOUR BLINDNESS IS A VISUAL IMPAIRMENT THAT AFFECTS A SIGNIFICANT PROPORTION OF THE POPULATION, WITH DIFFERENT TYPES OF COLOUR BLINDNESS PRESENTING UNIQUE CHALLENGES TO INDIVIDUALS ATTEMPTING TO INTERPRET DATA VISUALIZATIONS. THE DEVELOPMENT OF ACCESSIBLE AND COMPREHENSIBLE DATA VISUALIZATIONS IS CRUCIAL FOR ENSURING THAT INDIVIDUALS WITH COLOUR BLINDNESS CAN ENGAGE WITH AND UNDERSTAND DATA-DRIVEN INSIGHTS.



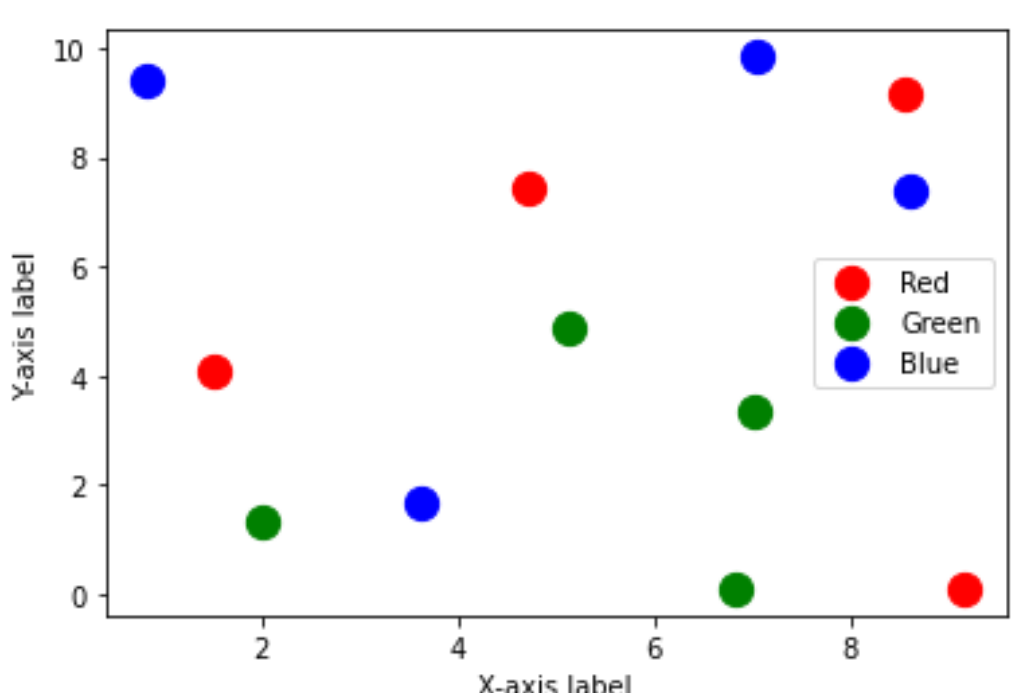
Normal Vision



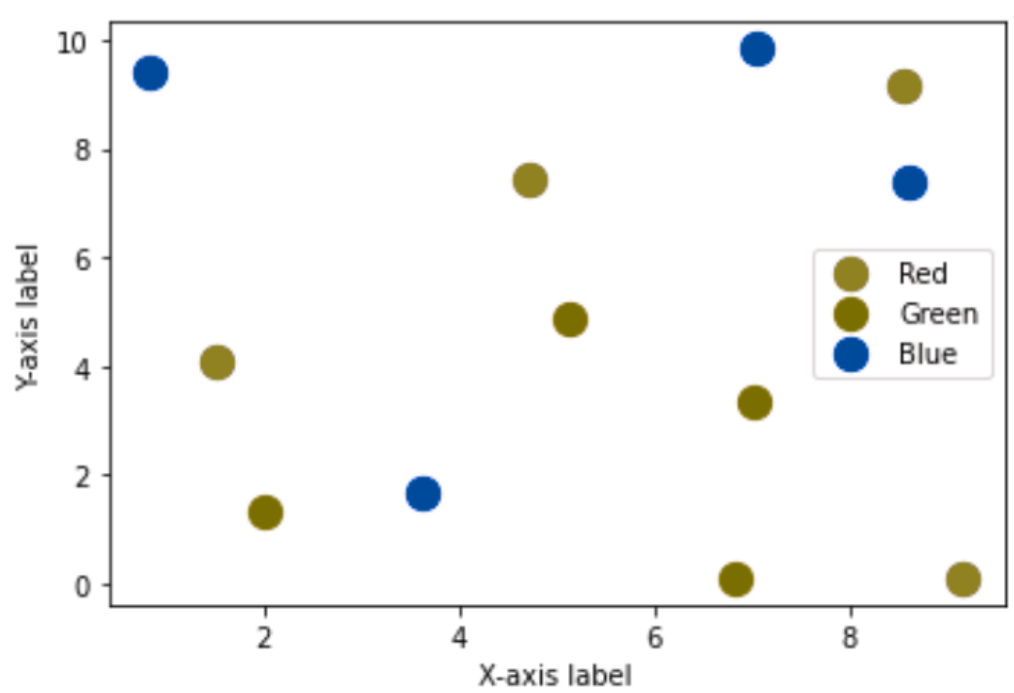
Blue Blind (Tritanopia)



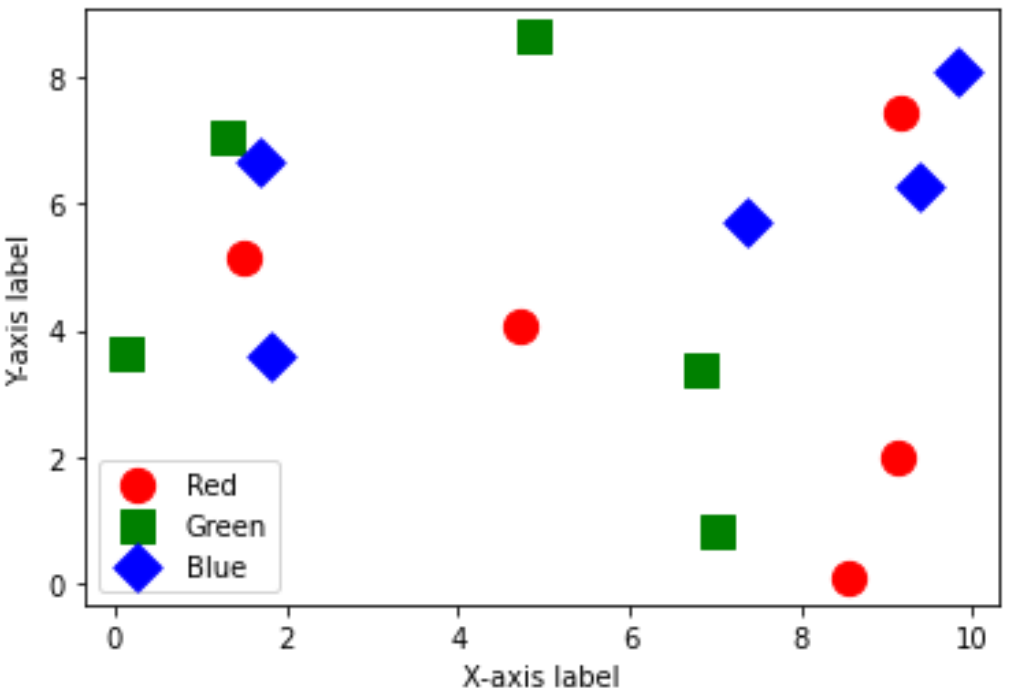
Green Blind (Deuteranopia)



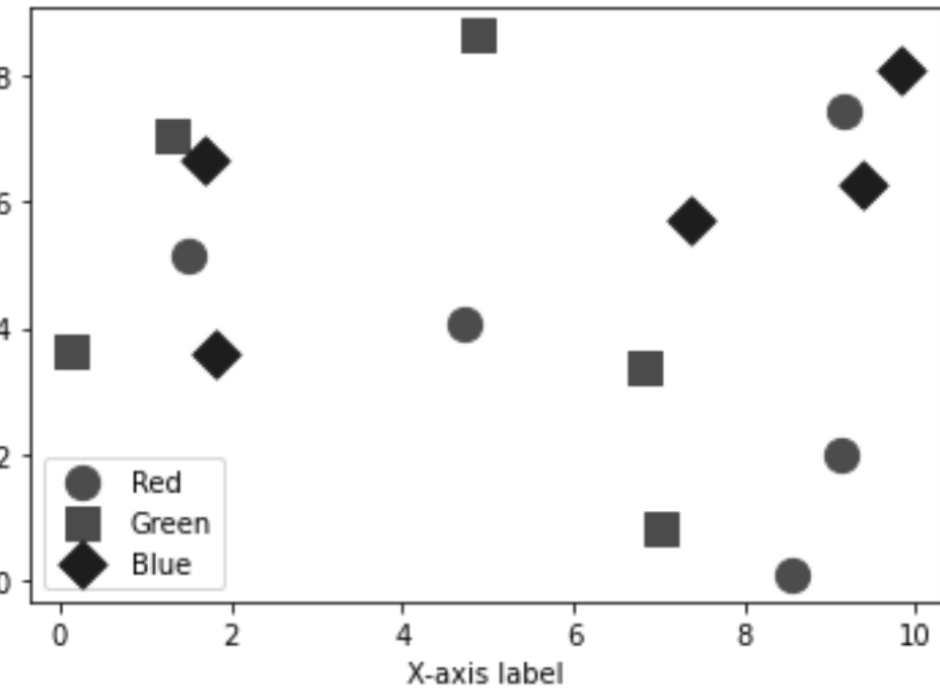
Normal Vision



Red Blind (Protanopia)



Correcting visualization



Monochromacy

Made using Coblis — colour Blindness Simulator & Python

DIFFERENT TYPES OF COLOUR BLINDNESS

Colour blindness is a visual impairment caused by a deficiency of certain colour-sensitive cells in the retina, leading to an altered perception of colour. There are multiple types of colour blindness, including protanopia, deuteranopia, and tritanopia, each resulting in a different colour confusion. For instance, **protanopia** impacts the distinction between red and green colours, while **deuteranopia** affects green and red colours. **Tritanopia**, in turn, impairs blue and yellow colour perception, and sometimes green shades. People with colour blindness can face challenges in various settings, especially in professions that rely on colour differentiation.

MALES ARE MORE LIKELY TO BE COLOUR BLIND

Males are more likely to be colour blind than females, with approximately **1 in 12 males** affected by some form of colour blindness. as opposed to **1 in 200 females**.

The genes responsible for colour vision are located on the X chromosome; males have only one copy of which, making them more susceptible to inheriting colour blindness.

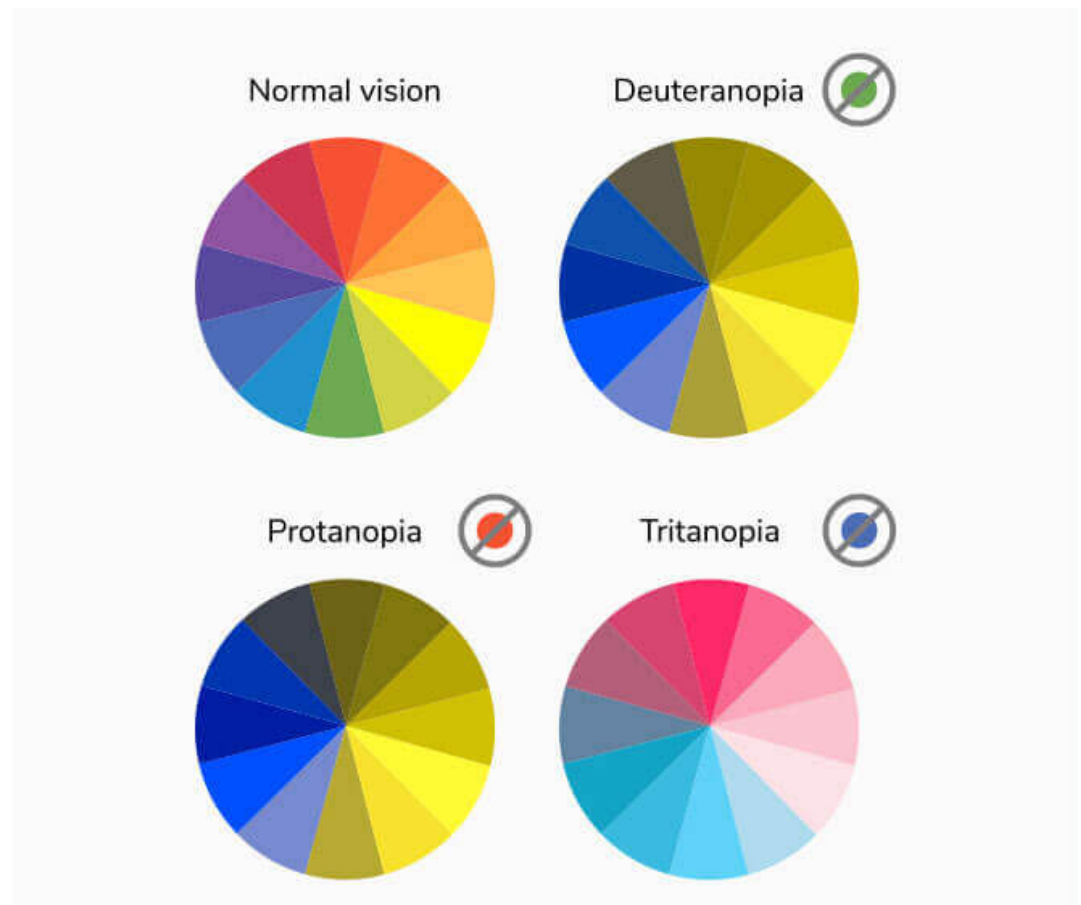
WHY IS IT NECESSARY TO MAKE COLOUR BLIND FRIENDLY VISUALIZATIONS?

Making colour blind friendly visualizations is necessary to ensure that individuals with colour vision deficiencies can understand and interpret data accurately, which is essential for **effective communication**; Ignoring the same can result in misinterpretation of data and exclusion of a significant portion of the population from important information. (**About 300 million people are colourblind**)

2.



Males are more likely to be colourblind



Different types of colour blindness & their respective colour wheels.
Image courtesy: venngage.com

3.

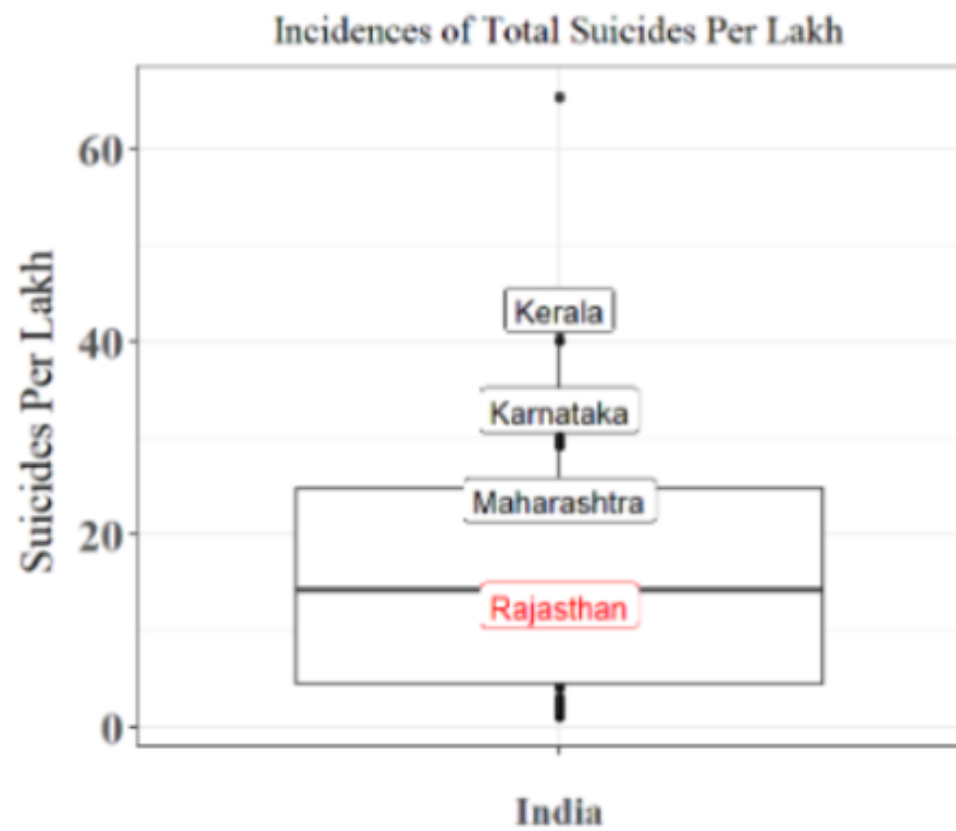
HOW COULD WE MAKE VISUALIZATIONS COLOURBLIND FRIENDLY



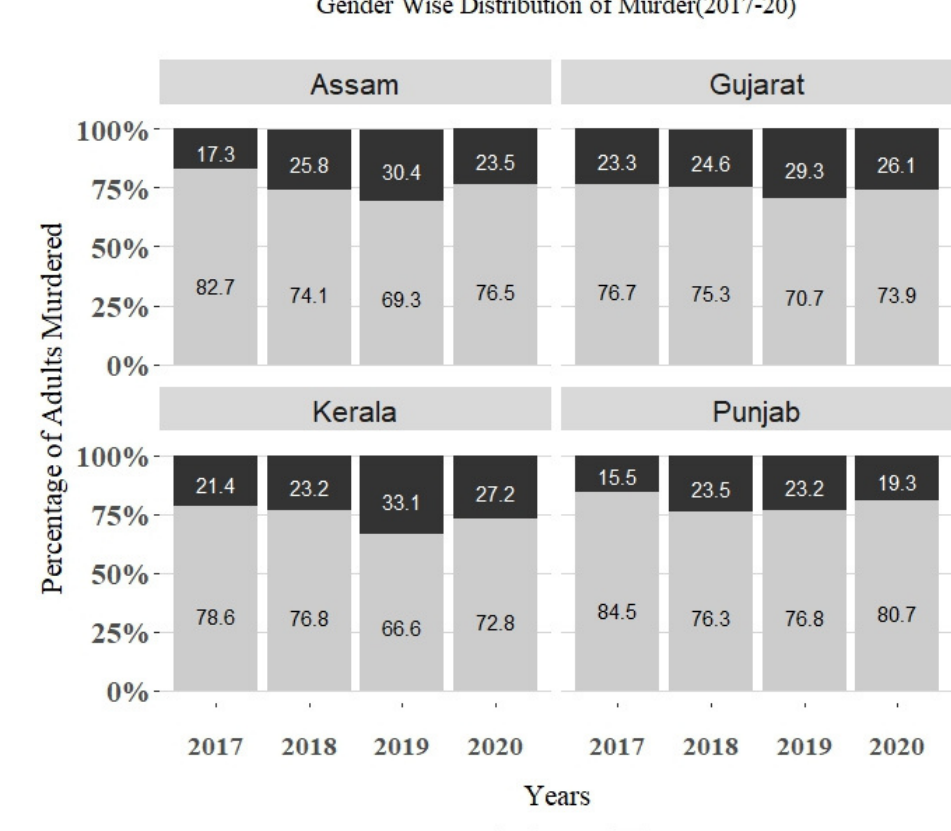
Monochromatic or Contrasting colours



Shapes & Patterns



Clean Labelling & Annotation



Faceting

HOW I ATTEMPT TO MAKE VISUALIZATIONS COLOURBLIND FRIENDLY

I use monochrome black and white themes alongside adding patterns to fill my graphs. Most often, I use red to highlight any relevant information.

You could use the ggthemes library in R and the hatch command in matplotlib in Python to add patterns to any visualization. Sometimes, monochrome palettes won't suffice, in that case there are plenty of colourblind friendly colour palettes online, my favorite is the Zissoul palette

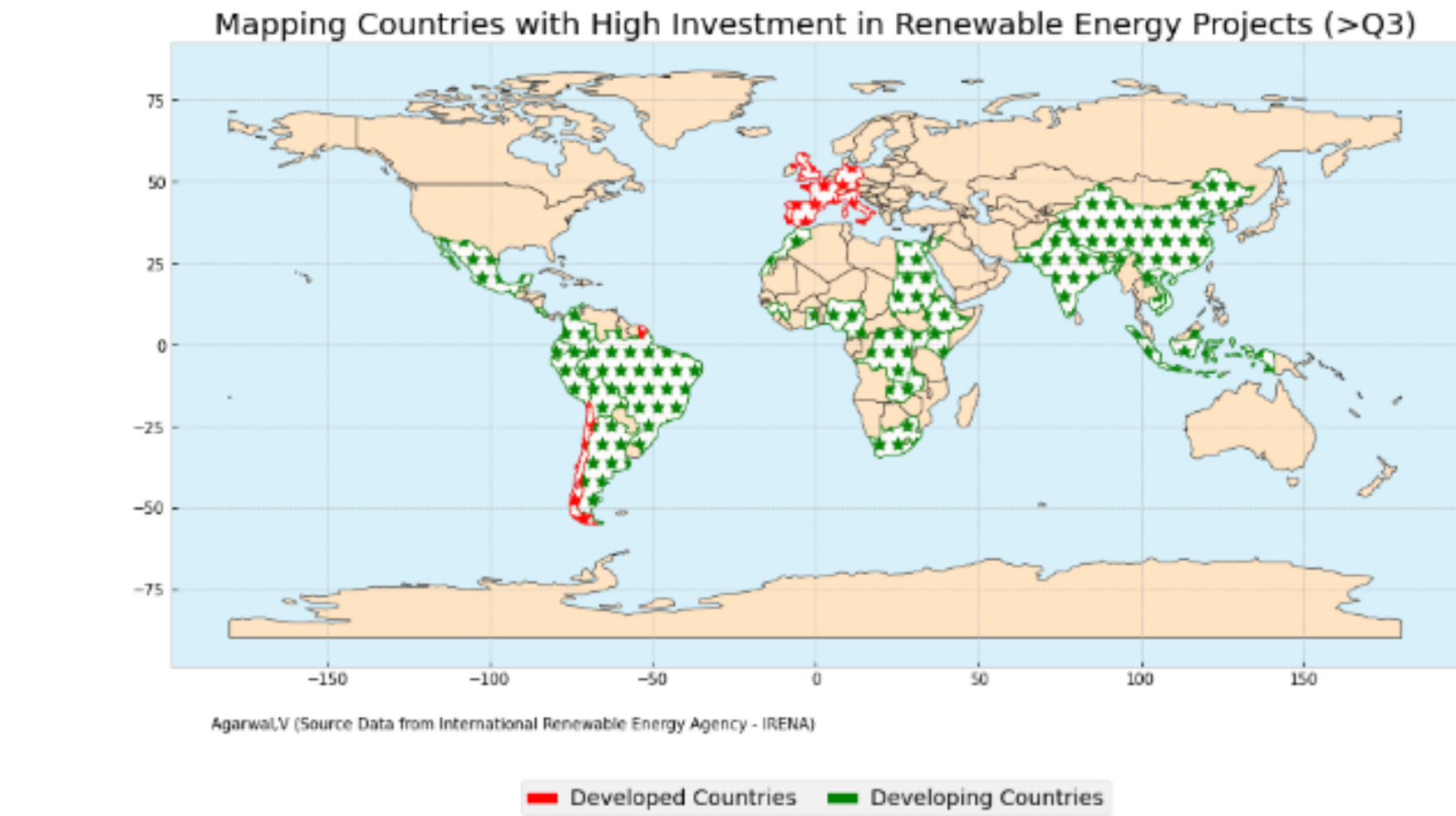


Normal Vision



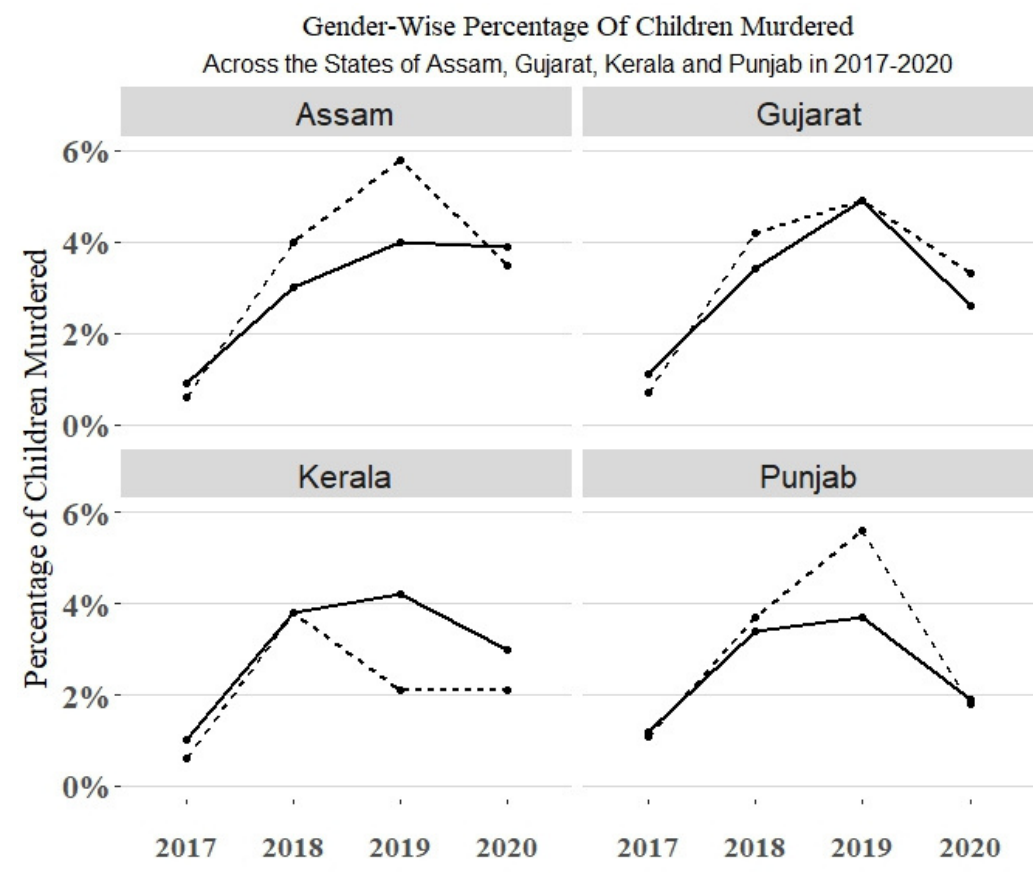
Blue Blind (Tritanopia)

Something as simple as being able to distinguish between the states categorised under northern or central parts of India would be difficult for an individual with tritanopia; adding patterns to maps make them accessible to all



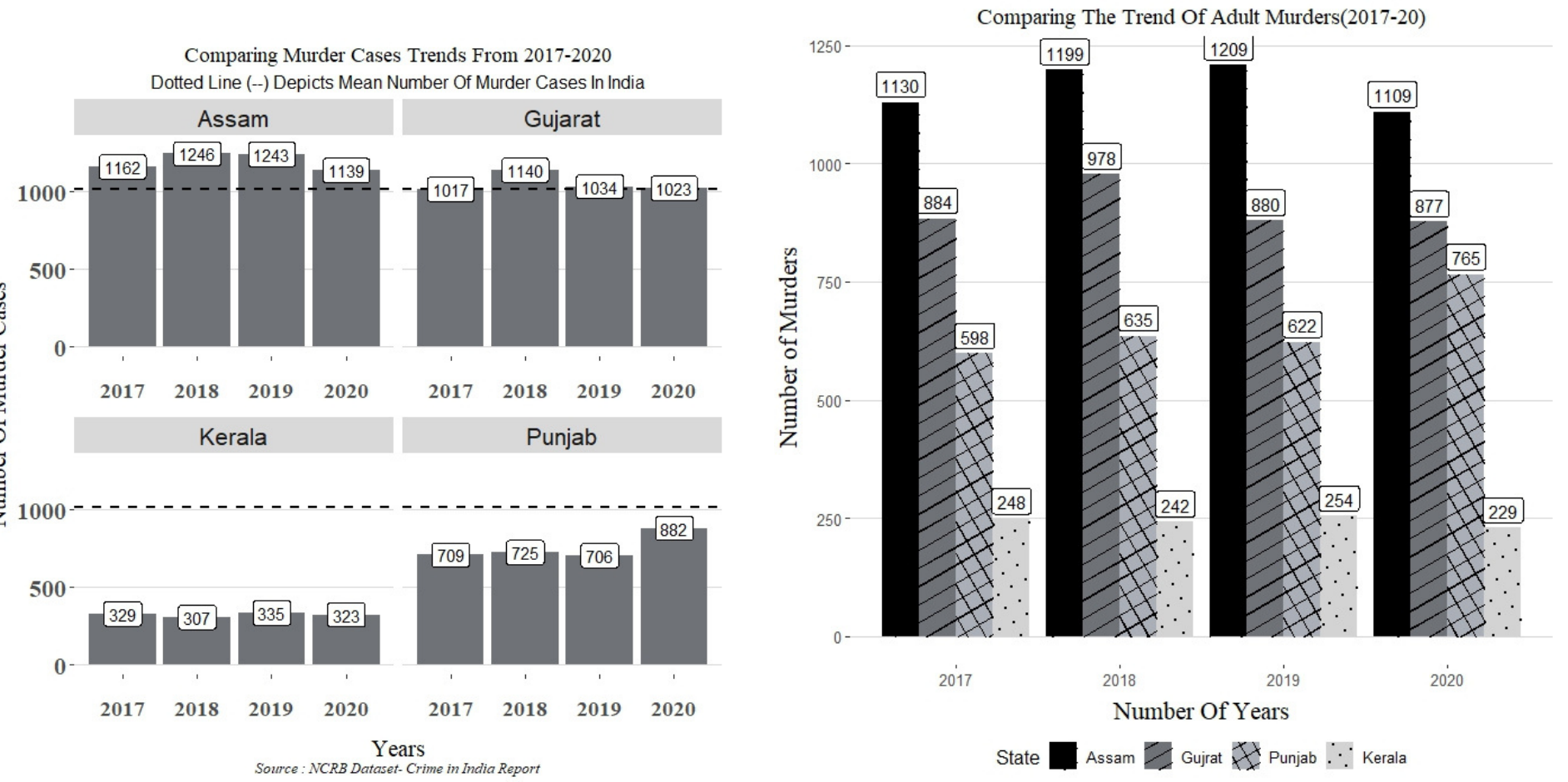
Choropleth maps colour code countries based on frequencies can be misleading and incomprehensible for the colourblind. Here is an alternative which uses patterns to highlight relevant information. Changing the palette is another alternative for continuous data

LINE GRAPHS



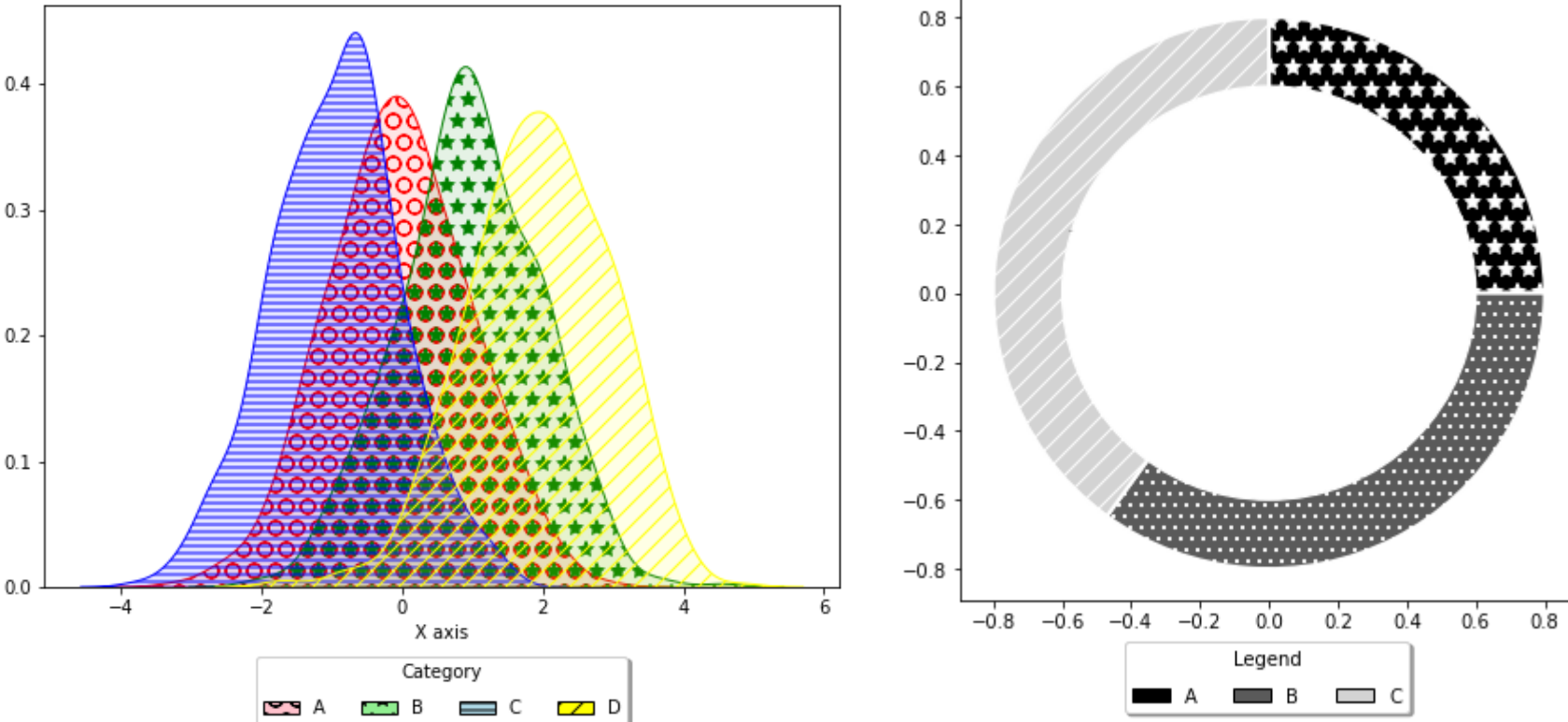
Using different types of lines (dotted or straight) and faceting to avoid use of colour to differentiate between categories, we could convert multiple line graphs colourblind friendly.

BAR GRAPHS



Both these graphs present the same data in two different colourblind friendly formats. The first one uses faceting, whereas, the second graph uses patterns. Adding labels to these graphs makes them more comprehensible to any audience.

DENSITY PLOTS AND PIE CHARTS



Any visualization can be made colour blind friendly; it is not necessary to make all visualizations grayscale. Colour palettes which are not colour blind friendly alongside patterns, clean labeling/ annotations or faceting, makes them colour blind friendly as well.

CONCLUSION

Through this poster presentation, I wish to convey the importance of color blind friendly visualizations and motivating my audience to generate visualizations which are colour blind friendly.